# Landau Volume 1

## 1 The equations of motion

### 1.1 Generalized Coordinates

Mechanics is interested in so-called "material-points", these are objects whose dynamics can be specified purely by their motion.

Positions of particles are defined by a radius vector r, which have components that are its Cartesian coordinates.
$r' = v$ this is velocity
$r'' = v' = a$ this is acceleration
For N particles we will need N radius vectors. If describing the movement in a 3D space, we will want 3N coordinates.

We have 3N degrees of freedom because this is the amount of information it takes to define the systems position coherently.

Unnecessary for DOF to be cartesian coordinates of particles, in other circumstances it may be more helpful to specify a different system such as one dependent on polar or spherical coordinates.

We define any set of s quantities $\sum_{j=0}^{s} q_j$ that completely define the position of a system with s DOF to be a generalized coordinate system, derivatives of $q_j$ are its generalized velocities.

Knowing the positions of a system does not strictly enable the prediction of its position, as you must also know the time evolution of those positions over infinitesimal periods of time. However, if these coordinates and velocities are specified, the state of the system is entirely determined.

Relations between position velocity and acceleration are called equations of motion.

**Principle of Least Action** Principle of Least Action / Hamilton's principle is the most general law governing mechanical systems. This allows for every mechanical system to be specified by a function of the form:

$$L(q, q', t)$$

If we take the system between two points $t_1, t_2$ and associate with it two sets of coordinates $q_1$ and $q_2$ then we have a condition of movement where the following integral takes the least possible value:

$$S = \int_{t_1}^{t_2} L(q, q', t)dt$$

The function L is called the Lagrangian and the integral is called the action. Lagrangian not containing higher derivatives is a consequence of position and velocity specifying the entire system.

We want to derive the diffeqs that will minimize this integral. Initially, assuming 1 DOF for simplicity so only one function $q(t)$ has to be determined:

Let $q = q(t)$ be the function for which S in minimized, so S is increased when $q(t) = q(t) + \delta q(t)$. $\delta q(t)$ is called a variation of $q(t)$. We say that $\delta q(t)$ is small everywhere from $t_1$ to $t_2$. Since for $t_1$ and $t_2$ we have $q_1$ and $q_2$ it follows that:

$$\delta q(t_1) = \delta q(t_2) = 0$$

What are we saying here? We are saying that the Varied path is equal to the true path plus the variation. We then can substitue our initial time, associated with the initial position $q_1$ into this expression: $q_1$, so we have $q_1 = q_1 + \delta q_1(t_1)$, we are able to say this because we know that since the variation is "small everywhere" its derivative is also well behaved, meaning that we would not leave $q_1$ with this subtle perturbation and this forces a condition where we know that after $t_1$ we will not radically leave $q_1$. We take this to be true by our definition of variation: $\delta q_1(t_1) = 0$, and note how it immediately leads to the well behavedness noted before. By similar reasoning on $t_2$ we obtain $\delta q_2(t_2) = 0$ and by the transitivity of equality: $\delta q_1(t_1) = \delta q_2(t_2)$

To find the change in S when q is replaced by $q + \delta q$ we find the area between $L(q + \delta q)$ and $L(q)$:

$$\int_{t_1}^{t_2} L(q + \delta q, q' + \delta q', t)dt - \int_{t_1}^{t_2} L(q, q', t)dt$$

We can intuitively think of the effect of variation as 'trapped' between the true path and the varied path. When we expand the difference in powers of $q'$ and $q$ the resulting dominant and significant terms are the lowest order ones:

Using the basic principle of $f(x + dx) \approx f(x) + f'(x)dx$,

$$L(q + \delta q, q' + \delta q', t) \approx L(q, q', t) + \frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial q'}\delta q'$$

We notice that subtracting this with $L(q, q', t)$ would result in:

$$\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial q'}\delta q'$$

Which can be interpreted as the sensitivity of the position and velocity multiplied by our perturbation of the position and the velocity. Landau seems to suggest that a subtraction of the Taylor series of both of these would yield similar results, as we can see our result as a first order term of a Taylor expansion.

For S to have a minimum, we need these terms (called first variation) to be equal to zero. The "principle of least action" is therefore written as:

$$\delta S = \delta \int_{t_1}^{t_2} L(q, q', t)dt = 0$$

or in our case:

$$\int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial q'}\delta q')dt = 0$$

We now have to use the fact that $\delta q' = d\delta q/dt$ and use integration by parts:

$$\delta S = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial q'}\delta q')dt = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q)dt + \int_{t_1}^{t_2} (\frac{\partial L}{\partial q'}\delta q')dt$$

By the earlier fact, we have:

$$\delta S = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial q'}\delta q')dt = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q)dt + \int_{t_1}^{t_2} (\frac{\partial L}{\partial q'}(d\delta q/dt)dt$$

Which equals;

$$\delta S = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial q'}\delta q')dt = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q)dt + \int_{t_1}^{t_2} (\frac{\partial L}{\partial q'}(d\delta q))$$

We let:

$$I = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q'}(d\delta q))$$

Recalling the integration by parts expression, $\int udv = uv - \int vdu$ we let $(u = \frac{\partial L}{\partial q'}) \implies (du = \frac{d}{dt}(\frac{\partial L}{\partial q'})dt)$ and $(dv = d\delta q) \implies (v = \delta q)$, substituting back we get:

$$I = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q'}(d\delta q)) = \left[\frac{\partial L}{\partial q'} \cdot \delta q\right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \delta q \frac{d}{dt}(\frac{\partial L}{\partial q'})dt)$$

Substituting this back into our expression from earlier, we get:

$$\delta S = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q)dt + \left[\left[\frac{\partial L}{\partial q'} \cdot \delta q\right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \delta q \frac{d}{dt}(\frac{\partial L}{\partial q'})dt)\right]$$

Using the earlier result that $\delta q(t_1) = \delta q(t_2) = 0$ we see the middle term drop out and we obtain:

$$\delta S = \int_{t_1}^{t_2} (\frac{\partial L}{\partial q}\delta q)dt - \int_{t_1}^{t_2} \delta q \frac{d}{dt}(\frac{\partial L}{\partial q'})dt = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt}(\frac{\partial L}{\partial q'})\right]\delta q dt$$

We can think of the variation now as acting similar to an expected or mean value for the function. Our derivation also aligns with what Landau has concluded. We then go to:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial q'}\right) - \frac{\partial L}{\partial q} = 0$$

But how do we justify this leap? By Landua's reasoning: we want an integral which must vanish for *all* values of $\delta q$ which can only happen if the integrand is also zero. But why?

Suppose that the integrand is nonzero, meaning that the result is some function of t, it is immediately clear why this will not be equal to zero for all values of t, since for $g(t) \neq 0$ the integral of $g(t)$ will result in a higher order term that will not strictly evaluate to 0, we also recognize that this much evaluate to 0 when multiplied by an arbitrary $\delta q(t)$.

Admitting that the integrand must be equal to 0, we take the earlier statement and manipulate it algebraically to get:

$$0 = \frac{\partial L}{\partial q} - \frac{d}{dt}\left(\frac{\partial L}{\partial q'}\right)$$

This matches Landau's derivation. When the system has more than one degree of freedom, we generalize this derivation by indexing the q' and q terms, resulting in s equations of the form:

$$0 = \frac{\partial L}{\partial q_j} - \frac{d}{dt}\left(\frac{\partial L}{\partial q'_j}\right) \quad (j = 1, 2, ..., s)$$

We vary these equations independently by the principle of least action. These equations are called Lagrange's equations. If the Lagrangian of a mechanical system is known, the above equations give the relationships between accelerations, velocities, positions; It can be equivalently said that they are the equations of motion for a system.

These equations constitute a set of s second order equations for s unknown functions $q_j(t)$. The general solution constants $2s$ arbitrary constants. To determine these constants and wholly define motion of the system as a result, we must know the initial conditions of the system, e.g all the initial coordinates and velocities.

If we let a mechanical system have distinct parts A and B, and treat them as closed so that they have lagrangians $L_A$ and $L_B$ respectively, the limit where the distance between them becomes arbitrarily large will reduce the coupling between them until eventually the lagrangians become separable such that:

$$\lim(L) = L_A + L_B$$

This means that the total lagrangian for two noninteracting systems is their sum. Take the earlier expression for system A;

$$\frac{d}{dt}\left(\frac{\partial L_{total}}{\partial q'_A}\right) - \frac{\partial L_{total}}{\partial q_A} = 0$$

However, we can use the relation that $L_{total} = L_A + L_B$ to rewrite the expression as:

$$\frac{d}{dt}\left(\frac{\partial L_A}{\partial q'_A} + \frac{\partial L_B}{\partial q'_A}\right) - \left(\frac{\partial L_A}{\partial q_A} + \frac{\partial L_B}{\partial q_A}\right)$$

But since A and B are completely disjoint, the derivatives of the Lagrangians of B with respect to the positions of A will be zero, which leads us to the form of the equation as it would exist for solely system A, exactly as we would expect.

The multiplication of the lagrangian of a mechanical system by an arbitrary constant has no effect on the equations of motion, this global scaling not changing the laws of the system can be justified by calculus in that if we are to take the action integral and multiply the lagrangian by an arbitrary constant:

$$S = \int_{t_1}^{t_2} c \cdot L(q, q', t)dt = c \int_{t_1}^{t_2} L(q, q', t)dt$$

, thereby not interfering with the integration, and actually propagating this factor so that we have $\delta S \to c \cdot \delta S$. It may seem that Lagrangians of different isolated systems may be multiplied by different arbitrary constants, but the additive property interferes with this. We may imaging this as the two systems A and B having locally defined scaling transformations by means of an arbitrary constant. We can intuitively imagine the lagrangian of system $A$ and system B multiplied by arbitrary constants as fibers over the entire system-space and say that this problem is equivalent to asking if there is a straightforward way of transporting between them. We have $L_A \to c_1 L_A$ and $L_B \to c_2 L_B$, letting them be completely distinct: $L_{total} = L_A c_1 + L_B c_2$, but this sum becomes indeterminate, because of this indefiniteness.

We could potentially ask some questions about if we could extend this to be complex valued naively in order to resolve this, in the manner of:

$$L_{total} = (L_A + c_1 i) + (L_B + c_2 i)$$

Though this would have a propagating impact by making the lagrangian complex valued and therefore the action. Landau links the arbitrariness of the lagrangian to the arbitrariness of its unit of measurement.

Another statement about the lagrangian should be made: Consider two lagrangians $L'(q, q', t)$ and $L(q, q', t)$ differing by the total derivative with respect to time of some function $f(q, t)$ of coordinates and time:

$$L'(q, q', t) = L(q, q', t) + \frac{d}{dt} f(q, t)$$

The action integral can be phrased as:

$$S' = \int_{t_1}^{t_2} L'(q, q', t)dt = \int_{t_1}^{t_2} L(q, q', t)dt + \int_{t_1}^{t_2} \frac{d}{dt} f(q, t)dt = S + f(q^{(2)}, t_2) - f(q^{(1)}, t_1)$$

Which is obtained by recognizing that the action S is contained within the action $S'$ and applying the fundamental theorem of calculus to the term which converts $S$ to $S'$. We also note that this results in $\delta S = 0 = \delta S'$ with no issues:

$$\delta S' = \delta\left[S + f(q^{(2)}, t_2) - f(q^{(1)}, t_1)\right]$$

So that:

$$\delta S' = \delta S + \delta f(q^{(2)}, t_2) - \delta f(q^{(1)}, t_1)$$

But because the endpoints are fixed as established earlier, their perturbation/variation is equal to 0, hence allowing us to conclude that $\delta S' = \delta S$